

A spatially explicit trait-based approach uncovers changes in assembly processes under warming

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Abstract

The re-assembly of plant communities during climate warming depends on several concurrent processes. Here, we present a novel framework that integrates spatially explicit sampling, plant trait information and a warming experiment to quantify shifts in these assembly processes. By accounting for spatial distance between individuals, our framework allows separation of potential signals of environmental filtering from those of different types of competition. When applied to an elevational transplant experiment in the French Alps, we found common signals of environmental filtering and competition in all communities. Signals of environmental filtering were generally stronger in alpine than in subalpine control communities, and warming reduced this filter. Competition signals depended on treatments and traits: Symmetrical competition was dominant in control and warmed alpine communities, while hierarchical competition was present in subalpine communities. Our study highlights how distance-dependent frameworks can contribute to a better understanding of transient re-assembly dynamics during environmental change.

KEYWORDS

community assembly, environmental filtering, hierarchical competition, reciprocal transplant, spatial associations, symmetric competition, warming experiment

INTRODUCTION

Climate change alters the composition and functioning of plant communities (Harrison, 2020; IPBES, 2019). It affects plant individuals directly but also indirectly through changes in their biotic neighbours: Winners will continue losing their less-adapted neighbours that decline in abundance or become extinct (Román-Palacios & Wiens, 2020; Urban, 2015), and new plant species will become neighbours due to asynchronous and species-specific range shifts (Alexander et al., 2015; Rumpf et al., 2018). The interplay between the direct effects of climate change and the induced changes in biotic interactions is thus the key to fully understand and predict

future plant communities' reshuffling. However, we are still far from being able to disentangle the signals of jointly acting environmental and biotic processes behind observed diversity patterns Münkemüller et al., 2020.

According to the original community assembly framework, environmental and biotic processes filter which species from a regional species pool can succeed in local communities (Keddy, 1992). These filters are expected to leave specific patterns in the functional trait structure of communities (Figure 1A; Keddy, 1992; McGill et al., 2006). On the one hand, environmental filtering should lead to trait convergence by selecting for the limited number of viable functional strategies Cornwell et al., 2006. On the other hand, when competition

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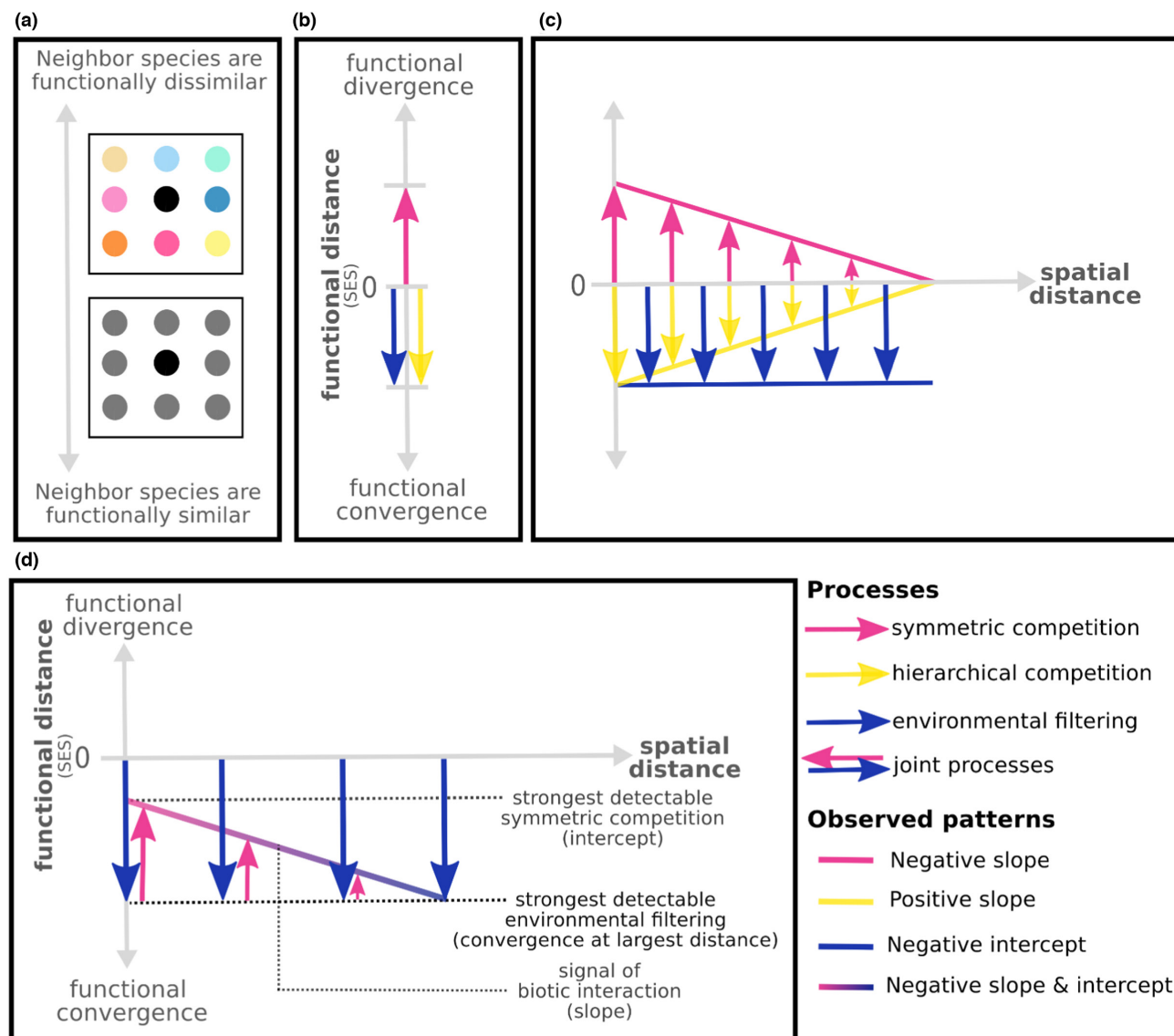


FIGURE 1 Spatially explicit trait-based framework to disentangle community assembly processes. In all the conceptual diagrams, the x-axis indicates the spatial distance and the y-axis indicates the functional distance between a focal individual and its neighbours standardized by randomizations (i.e., SES, see Equation 3). A functional distance below (above) 0 is a tendency towards functional convergence (divergence). (a) By observing different functional diversity patterns in a community, we can understand processes behind community assembly. As the focal and neighbour species become functionally less (more) similar, the functional distance between the focal and its neighbours increases (decreases). (b) The original framework of community assembly aims to disentangle processes behind functional patterns by studying only the functional distance between focals and their neighbours. This framework does not integrate spatial distances and thus cannot disentangle simultaneously acting processes. (c) In a spatial distance-based framework, the signal of environmental filtering remains constant with increasing spatial distance, while the effect of biotic interaction is fading away. (d) An example of a pattern when different processes are in interplay (here environmental filtering and symmetric competition characterized by limiting similarity) but patterns can be disentangled by distance-dependent analyses. Environmental filtering is constantly strong but the closer individuals the stronger the effect of limiting similarity. We used three indices to summarize these regressions: (i) a significant slope indicates a signal of potential biotic interactions, (ii) the intercept represents the functional diversity estimate at the smallest possible distance between neighbours where the effect of competition is strongest (i.e., stronger symmetric competition leads to stronger limiting similarity) and (iii) the predicted value at the largest neighbourhood distance is expected to show the strongest effect of environmental filtering as competition should be weakest here.

increases with functional similarity (i.e., symmetric competition) we should observe that trait divergence as coexisting plants must find distinct functional strategies to reduce the competition for limited resources (e.g., water, nutrients and light; MacArthur & Levins, 1967). For plants, a multitude of processes exist that may lead to limiting similarity. For example, a required resource

like nitrogen exists in different forms in the soil. Some plant species have a root system (e.g., with thinner and strongly branching roots) well equipped for direct access to the inorganic nitrogen pool while others with opposite traits invest in the collaboration with mycorrhizae, the indirect access to the organic nitrogen pool Weigelt et al., 2021. Similarly, trade-offs between acquisition

strategies are important: A trade-off between light acquisition and biomass allocation strategies allows the coexistence of tall and short species in herbaceous communities (Hirose & Werger, 1995) and a trade-off between light and nutrient acquisition strategies can lead to divergent patterns of leaf nitrogen content (Hirose & Werger, 1994).

Despite its vast applications, the original community assembly framework was criticized early on. First, competition does not always lead to limiting similarity but can as well lead to convergent traits due to hierarchical competition, when certain functional strategies have the ultimate competitive advantage while others are excluded. Such trait hierarchies are shown to be established in the case of competition for light (Kunstler et al., 2012; Mayfield & Levine, 2010). Second, the framework is insufficient to reveal simultaneously acting assembly processes, which, in the extreme case, may lead to random-looking patterns Münkemüller et al., 2020. Third, assembly processes depend on spatial scales and different processes are likely to act at different scales. While it is difficult to identify the most relevant scales, different scale choices can lead to contradictory results (Cavender-Bares et al., 2006; Chalmandrier et al., 2017; Münkemüller et al., 2012). For example, biotic filters may be overlooked if distances between studied individuals are too large Blanchet et al., 2020. Despite the awareness that scale can be the key in revealing community assembly rules, attempts to develop a spatially explicit approach are so far limited to either ad hoc manipulations of the dataset and/or species pool (Chalmandrier et al., 2017; de Bello et al., 2013), comparisons of only two opposing scales (Carboni et al., 2013; Cavender-Bares et al., 2006) or case-specific comparisons that do not provide a stringent spatially explicit framework Trisos et al., 2014; Xu et al., 2021. If relevant scales for environmental and biotic filters are unknown, we require a continuous spatially explicit sampling protocol and trait-based diversity analysis to disentangle assembly processes.

Here, we developed such a spatially continuous trait-based framework of community assembly (Box 1). By studying the relationship between spatial and functional distances between focal and neighbour individuals, we can detect the signals of dominant processes acting at community scale and can disentangle signals of environmental filtering from those of biotic interactions and symmetric competition from hierarchical competition. We applied our framework to a 2-year transplant experiment in mountain grasslands where both environmental filtering and competition filters are expected to act in concert Scherrer et al., 2019 and their relative strengths should depend on the severity of the environmental stress (Bertness & Callaway, 1994; Louthan et al., 2015; Sundqvist et al., 2013). We tested the following hypotheses using functional traits related to stress tolerance and competitive ability:

(H1) Environmental filtering is stronger in alpine than in subalpine grasslands due to more severe alpine conditions (i.e., colder temperatures and shorter growing seasons).

(H2) Competition is stronger in subalpine than in alpine grasslands due to more benign conditions and higher productivity.

(H3) With climate warming, the strength of environmental filtering weakens and the strength of biotic filtering strengthens. As weak or temporally changing environmental and biotic filters may rather induce abundance changes than local extinctions or colonizations, we expect that all patterns are stronger when they are not only based on presence-absences but are weighted by local abundances (Hu et al., 2021; Wang et al., 2021). This should be the case for the transplanted communities as they are still adjusting to new conditions.

METHODS

Experimental design

The reciprocal transplant experiment was conducted in the French Alps (Col du Lautaret—45°02'04" N 06°24'18" E) in September 2016 Bektaş et al., 2021. The subalpine site "Lautaret" (45°04'00" N 06°41'90" E at 1920m altitude) and the alpine site "Galibier" (45°05'44" N 06°40'06" E at 2450m altitude) are close to each other (~2km airline distance) and have a similar orientation (South-East at alpine and South-South-West at subalpine site), bedrock (base-riched flysch) and soil (dystric cambisols). In 2018, the year of data collection, average annual soil temperature at the subalpine site was 2.9°C warmer than at the alpine site, average July soil temperature was 2.8°C warmer and snow free season was 48 days longer. At the alpine site, communities were dominated by *Trifolium alpinum*, *Potentilla aurea*, *Plantago alpina*, *Carex sempervirens* subsp. *sempervirens* and *Poa alpina*. At the subalpine site, communities were dominated by *Patzkea paniculata* subsp. *paniculata*, *Festuca nigrescens*, *Carex sempervirens* subsp. *sempervirens*, *Festuca laevigata* and *Plantago maritima* subsp. *serpentina*. In 2016, to simulate the effect of warming on alpine grasslands, we transplanted 10 replicates of 4 m² plots of vegetation with at least 20 cm of their intact belowground parts and the surrounding soils from the alpine to the subalpine site (AlpineWarmed plots). For the transport, each plot was cut in 4 × 1 m² subplots and then reassembled in their original composition when re-planted. To control for the transplantation stress, we also horizontally transplanted 10 intact 1 m² plots at both the subalpine and the alpine sites (SubalpineControl and AlpineControl plots respectively). Within the same site, control and treatment plots are in close proximity within a few meters. Around the plots, we placed water permeable root barriers to avoid root ingrowth from the natural vegetation around the plots. Cattle and sheep were excluded

BOX 1 A spatially explicit trait-based community assembly framework.

We built our framework on two basic assumptions: (i) Environmental filtering is constant in small and homogeneous community plots (i.e., is distance independent); (ii) Interaction strength between plant individuals decreases with spatial distance (i.e., is distance dependent). We propose to quantify variation in functional trait distances between a focal individual and its neighbours along increasing neighbourhood distances (Thuiller et al., 2010). We use standardized effect sizes (SES, i.e., standardized by means and standard deviations of randomized communities, see Equation 3) as functional distances such that positive values indicate trait divergence, negative values trait convergence and values around zero random patterns (Gotelli & Graves, 1996). Plotting the resulting trait distances against spatial distances helps to disentangle jointly acting processes. A negative intercept and non-significant slope may reflect a signal of strong environmental filtering and no biotic interaction (Figure 1B, blue line). A significant slope may indicate a signal of any kind of biotic interaction (Figure 1B, pink and yellow line). In the case of competition, we can tease apart hierarchical vs. symmetric competition. A negative slope may indicate symmetric competition because the closer the individuals are, the more they limit the functional similarity of the others (Figure 1B, pink line). A positive slope may indicate hierarchical competition because the closer the individuals are, the more they are functionally clustered around the most competitive strategy (Figure 1B, yellow line). However, environmental filtering and biotic interactions can act jointly. For example, the interplay of a strong environmental filter and symmetric competition may result in a negative slope and mostly trait convergence (Figure 1C). At the smallest spatial distance, competition is strongest. While the simultaneously acting environmental filter pulls diversity to convergence, functional similarity is limited to the value at which species can still coexist under full competition. If species get any more similar, one will outcompete the other. Thus, the intercept in Figure 1C. quantifies the strongest detectable symmetric competition as stronger symmetric competition limits similarity more than weaker symmetric competition or—in other words—the intercept increases with increasing competition. Furthermore, at spatial distances that are too large for biotic interactions, the environmental filter acts alone. Convergent patterns at this spatial distance thus quantify the strength of environmental filtering. The question of how large this spatial distance needs to be can be answered by expert knowledge or by observing how convergence responds to increasing spatial distances and at which spatial distance it reaches a plateau. Although the intercept and the diversity at the largest measured spatial distance do not necessarily reflect the sheer strength of each assembly process, they are useful to compare relative strengths of filters between communities with different environmental conditions or climate change treatments.

Note that competition and facilitation signals cannot be easily disentangled with this framework because facilitation is also distance-dependent and can cause both convergence and divergence (Münkemüller et al., 2020). Additional assumptions and analyses are required to identify facilitation. In this study, we performed an additional spatial aggregation analysis that suggested that signals of facilitation are absent (Appendix S3). Moreover, if the assumption of homogeneous plots does not hold, a positive slope between spatial and functional distances could also signal micro-scale environmental filtering (Conti et al., 2017; Willis et al., 2009; Xu et al., 2021).

In this specific application, we focused on detecting dominant processes acting at the scale of a diverse community. It is possible that processes important only for a subpart of the community or jointly acting processes with weaker effects on diversity patterns are overlooked (Chalmandrier et al., 2022). In that particular case, our framework is flexible to be applied to subparts of the communities to reveal processes at finer scales. In addition, more refined null models in the calculation of SES values can be used to remove a pattern expected from a given process and to test if, after this, there is an additional signal in the data that could be linked to additionally acting but weaker processes (Hardy, 2008). Finally, other types of functional relationships between traits and performance (i.e., unimodal functions) can be explored through the use of specific indices (e.g., trait differences for trait hierarchies, Kunstler et al., 2012) and be interpreted in the light of their spatial distance dependency following our framework.

by electric fences and grazing was not simulated. There were no soil disturbances due to burrowing animals or erosion due to slope. More details on the experimental

design and the changes in phenology and productivity of communities due to experimental warming can be found in Bektaş et al. (2021).

Vegetation surveys

At the peak of the plant growth (July 2018), we identified and counted all species that touched a 4 mm metal stick (here after, pinpoint) at equally spaced points (~10 cm distance) in both the treatment (400 points in total for each repetition) and control (100 points in total for each repetition) plots. Species that could be identified only at genus level were excluded from the further analysis (5% of 15,726 identified individuals). The pinpoint counts were measured on a square grid which allowed us to know the exact position of pinpoints.

Defining focal and neighbour species

All identified and counted species at a pinpoint were considered as focal individuals (AlpineControl: 2267, SubalpineControl: 3109, AlpineWarmed: 9198). For each focal individual, we constructed four different neighbourhoods at different distances to the focal, starting at 10 cm (i.e., individuals touching the focal pinpoint and pinpoints at 10 cm distance) and then increasing distances (i.e., the 20 cm neighbourhood ring consisted of the individuals touching pinpoints at >10 cm and ≤30 cm and so forth). Pre-analyses showed that the 40 cm neighbourhood ring was a good compromise to identify environmental filtering only (i.e., competition signals were sufficiently reduced and there were enough focal pinpoints for which the most distant neighbourhood was still in the experimental plots [Figure S12]). Conspecifics were taken out from a focal's neighbourhood as they could belong to the same individual if touching the same pinpoint or clones if in close proximity. Since the methods we used may be sensitive to sample sizes, we subsampled 20 individuals 20 times with replacement within each neighbourhood and made all calculations within these sub-neighbourhoods. We chose the number of individuals to subsample so that we would not lose more than 15% of the pinpoints at the smallest neighbourhood in

any of the treatments. The abundances of the neighbours within sub-neighbourhoods were estimated as the total pinpoint touches within the sub-neighbourhood (i.e., all touches at all pinpoints in this sub-neighbourhood). We only included focal-neighbourhood combinations for which more than 80% of the neighbours had functional trait information and for which the entire neighbourhood was within the grid. While the number of focal-neighbourhood combinations decreased with increasing neighbourhood distance this way, these adjustments did not change the overall shape of the trait distributions, and thus, the remaining species represented well the trait space at each neighbourhood distance (Figure S1).

Regional species pool

We extended our regional species pool beyond the list of species in the experiment to be able to account for all species in the catchment area and thus all species available to enter the local communities Carstensen et al., 2013. The regional species pool (787 species) included species from (i) our experimental sites, (ii) botanical surveys from the long-term monitoring program ORCHAMP along the elevational gradient of the transplant experiment (i.e., from 1920 to 2700 m at five different sites, <https://orchamp.osug.fr/sites/5>), and (iii) botanical surveys (i.e., 747,387 observation points) from the Alpine National Botanical Conservatory (CBNA; <http://www.cbn-alpin.fr/>) taken within a 5-km buffer around the experimental sites. As our experimental sites are grasslands, we excluded shrubs, trees and understory herbs from the regional pool to avoid a signal of an additional habitat filtering process.

Interspecific functional trait data

We considered five different traits that reflect different functional plant strategies and competitive strength within communities (Table 1). We also considered an

TABLE 1 Selected plant functional traits.

Trait	Function (high trait values result in ...)	Species interactions (high trait values result in ...)
Vegetative height	More photosynthetic growth, higher contribution to community productivity	Stronger competition for acquiring light and creating shade to neighbours
Specific leaf area (SLA)	More photosynthetic growth, higher contribution to community productivity, more intolerance to harsh environmental conditions	Stronger competition for acquiring light and creating shade to neighbours
Leaf dry matter content (LDMC)	More resistance to harsh environmental conditions notably drought and disturbance	Slower resource acquisition
Leaf carbon nitrogen ratio (CN)	More resistance to infertile soils	Slower resource acquisition
Seed mass	More intolerance to harsh conditions in the soil	Higher competitive advantage in establishment

Note: Synthesized from Martínez-Almoyna et al. (2020), Saatkamp et al. (2019) and Venn et al. (2014).

integrative trait of vegetative height, specific leaf area (SLA) and leaf dry matter content (LDMC). These three traits represented the multidimensional trait space of the species pool well (Figure S4). We log-transformed and then we transform the trait values so that they follow a normal distribution with the Gaussianize function for skewed heavy tails in LambertW package (Goerg, 2022; Figure S5). We extracted the interspecific functional trait data from our in-house trait measurements (Brun et al., 2022; Thuiller et al., 2014) available in the Androsace database (<https://originalps.osug.fr/>) and complemented the missing data for all the traits in Table 1 from the TRY database Kattge et al., 2020 and the Tundra Trait Team database Bjorkman, Myers-Smith, Elmendorf, Normand, Thomas, et al., 2018. We detail the data selection and manipulation processes from these databases in Appendix S2.

Species with no trait information at all were removed (3.8% of the regional pool), remaining species formed the updated species pool. The integrative trait was only calculated for those species for which all three traits were available (619 species, 82% of the updated species pool). Leaf CN ratio and seed mass were calculated based on 525 and 476 species (69% and 63% of the updated species pool) respectively. Based on species relative abundances, the coverage of the trait data in the experimental communities was above 97% for all traits (Figure S3).

Functional distances

Some research suggests that environmental filters are more important to determine species presence or absence, while biotic interactions rather affect species' abundances Boulangeat et al., 2012. In addition, transient dynamics of communities under change may first respond by abundance changes and only in the long-term by extinctions and colonization events. To account for this, we calculated two metrics, an abundance-weighted (Equation 1) and an unweighted (Equation 2) mean functional distance between a focal and its neighbours accordingly Thuiller et al., 2010:

$$\text{WMDN} = \frac{\sum_{i=1}^N w_i \times d_i}{\sum_{i=1}^N w_i} \quad (1)$$

$$\text{MDN} = \frac{\sum_{i=1}^N d_i}{N} \quad (2)$$

where N is the number of neighbour species, w_i is the relative abundance of the neighbour species i , and d_i is the functional distance between the focal and the neighbours i . The functional distance was calculated as the Euclidean distance between focals and their neighbours for each single trait and for the integrated trait.

Null model

Our null hypothesis was that “observed neighbours of a species are randomly selected from the regional species pool.” For this, we randomized the pinpoint x species abundance matrix used to identify neighbourhoods by randomly switching observed species names with species names from the species pool. That way, species traits are taken at random from the species pool without changing species numbers and abundances. We then calculated for all focals and all sub-neighbourhoods functional distances for observed and randomized data. As these functional distances were calculated at 20 sub-neighbourhoods, we later took the median to have a single value for each focal species at each neighbourhood distance for observed and randomized data. We calculated the standardized effect size (SES) for each individual and at each neighbourhood distance based on median values as follows:

$$\text{SES} = \frac{\text{Metric}_{\text{observed}} - \text{mean}(\text{Metric}_{\text{Null}})}{\text{sd}(\text{Metric}_{\text{Null}})} \quad (3)$$

Negative (positive) values of SES indicate that focal individuals are functionally more similar (dissimilar) to their neighbours than expected by chance (i.e., functional convergence vs. functional divergence).

Statistical analyses

First, we tested for significant patterns of functional convergence and divergence of focals independently for each neighbourhood distance and for all metrics, traits and experimental treatments. We used the two-sided Wilcoxon signed-rank test (*wilcoxon.test* in R) to test if the focal individuals' SES values were significantly different from zero in order to control for multiple testing Bernard-Verdier et al., 2012.

Independently for all metrics, traits and experimental treatments, we built linear mixed-effects regression models with spatial distances as explanatory variables, SES functional distances as responses and repetition plots as random effects (*lme4* package in R). In functional distance calculations, neighbourhoods overlapped. We corrected for this by weighing each observation by its pinpoints' nonoverlap ratio within its plot. We calculated the latter as follows:

$$1 - \frac{\text{Overlap}_{ij}}{\sum_{i=1}^N \text{Overlap}_{ij}} \quad (4)$$

where Overlap is the number of overlaps of a pinpoint i with any of the neighbourhoods in repetition plot j and N is the number of pinpoints within the same repetition plot.

We calculated predictions with the *lmerTest* and the *ggeffects* package (Kuznetsova et al., 2017; Lüdtke, 2018). We used slope, intercept and the predicted value at the largest neighbourhood distance to summarize these regressions and to interpret them according to our framework (Figure 1C).

We did not rely on significance tests of the regression models as measured functional distance values were not necessarily normally distributed and independent from each other. We thus repeated the entire analyses on 100 random species compositions, using the same null model as described above. We then compared intercepts and slopes of these “test” data with our observed data and based the significance test on this comparison (Figures S8, S9 and S10). We report deviations from random patterns as significant if values are lower or higher than 95% of random values (i.e., two-sided null hypotheses) and as strong tendencies if they are higher than 95% of random values or lower than 95% of random values (i.e., one-sided null hypotheses).

We detected the potential signal of any biotic interaction through a significant slope between spatial and functional distances. We further checked if there is a potential signal of facilitation in our system with an additional spatial aggregation analysis (Appendix S3; Figure S6). Once we confirmed that a signal of facilitation is absent, we identified the potential processes behind competition at a finer scale: a negative slope can signal symmetric competition and a positive slope can signal hierarchical competition. We quantified the relative strength of competition through the intercept as it estimates the functional distance at the smallest spatial distance between neighbours where the effect of competition is expected to be strongest (see Box 1 and Figure 1).

To test H1 and H2, we compared the above-mentioned metrics between AlpineControl and SubalpineControl communities. To test H3, we compared AlpineWarmed communities to AlpineControl and SubalpineControl Bektaş et al., 2021 with the expectation that warmed communities showed a transient response between AlpineControl and SubalpineControl communities. Moreover, to better understand how abundances influence results, we compared regression results for WMDN (abundance weighted) and MDN (presence-absence based). We conducted all the statistical analyses in R 4.2.2 (R Core Team, 2022).

RESULTS

Comparison of alpine and subalpine control plots

At all neighbourhood distances, SES values (WMDN and MDN) indicated that focals were functionally more similar to their neighbours than expected by chance (i.e.,

significant functional trait convergence; Figure S7) with few exceptions for SubalpineControl in SLA and leaf CN ratio (Figure S7).

AlpineControl and SubalpineControl communities showed different signals of competition (Figure 2). AlpineControl communities had significant negative slopes in the integrated trait, vegetative height and seed mass. SubalpineControl communities had a significant negative slope in LDMC and a positive tendency in SLA and leaf CN ratio (Figure 2). In sum, signals of symmetric competition signals were prevailing in AlpineControl communities, while SubalpineControl communities showed signals of both hierarchical and symmetric competition (Figure 1).

Environmental filtering was observed for all traits with the exceptions of SLA and leaf CN ratio. It was generally stronger in AlpineControl than in SubalpineControl communities. Only the environmental filtering of LDMC had a strong tendency in SubalpineControl communities (Figure 2).

Overall, signals of jointly acting assembly processes were present in alpine communities for the integrated trait, vegetative height and seed mass (significant, Figure 2) and in subalpine communities only for LDMC (significant environmental filter and symmetric competition, Figure 2).

Community reassembly under climate warming

AlpineWarmed communities showed negative slopes as signals of symmetric competition in the integrated trait, vegetative height, seed mass (significant, Figure 2) and LDMC (strong tendency, Figure 2). Signals of environmental filtering were present for the integrated trait, vegetative height, LDMC and seed mass (all significant, Figure 2). In fact, AlpineWarmed communities were in a transient state where observed relationships between functional and spatial distances were in between the two controls with the exception of LDMC and seed mass (Figure 2).

Warming effect (i.e., comparing AlpineControl to AlpineWarmed) increased limiting similarity of symmetric competition for the integrated trait and vegetative height (i.e., higher intercept, Box 1, Figure 3), decreased limiting similarity of symmetric competition for seed mass (i.e., lower intercept, Box 1, Figure 3) and added a tendency of symmetric competition for LDMC (Figure 2). Thus, overall there was a light increase in competition signal (except for seed mass) but overall effects were weak (Figure 3). In contrast, environmental filtering signals were strongly released with warming with the exception of leaf dry matter content (Figure 3).

Changes in local abundances pronounce warming effects

Including abundance information (WMDN) or not (MDN) to calculate functional distances did not

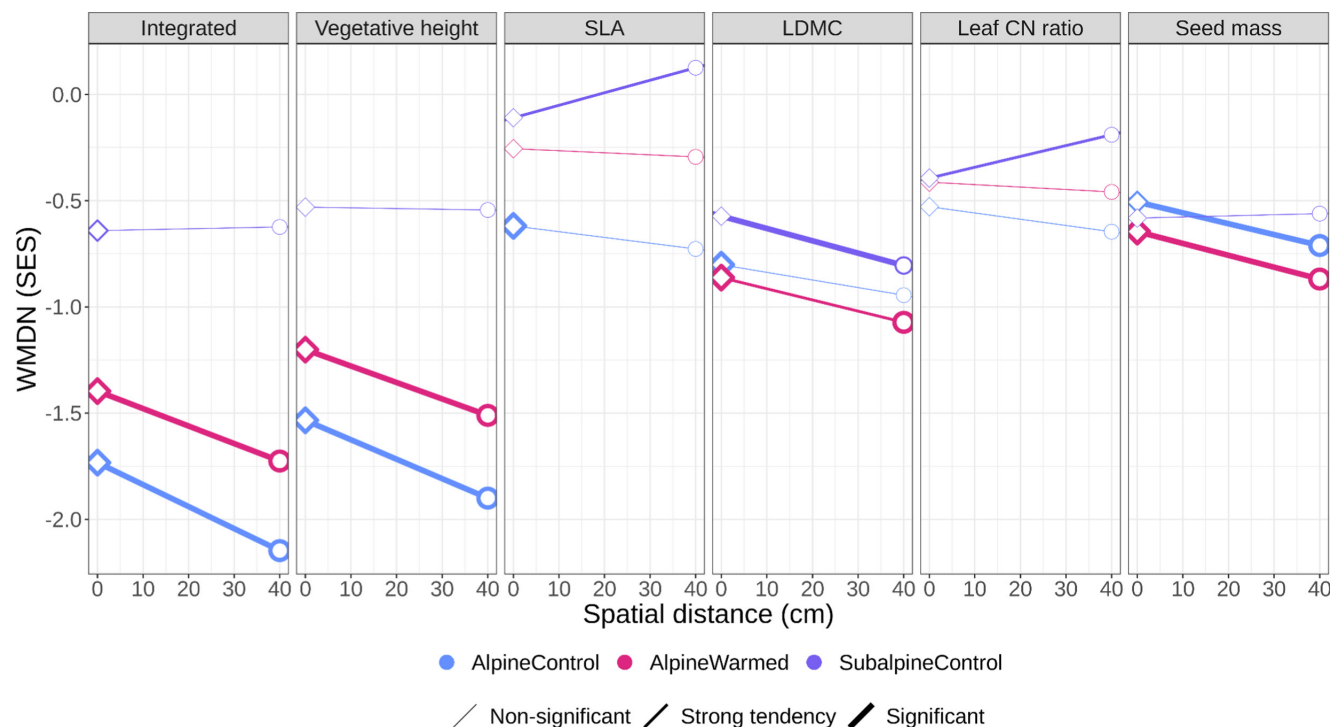


FIGURE 2 Changes in SES of functional distances (WMDN) with increasing spatial distances as predicted by the linear mixed-effect regressions. A significant slope is a signal of biotic interaction. A significant intercept is a signal of environmental filtering (diamond). The intercept quantifies the strongest detectable biotic interaction if there is any. The predicted value at the largest distance is the strongest detectable environmental filtering (circle). We report deviations from random patterns as significant if values are lower or higher than 95% of random values (i.e., two-sided null hypotheses) and as strong tendencies if they are higher than 95% of random values or lower than 95% of random values (i.e., one-sided null hypotheses). Integrated: combination of vegetative height, SLA & LDMC, SLA: specific leaf area, LDMC: leaf dry matter content, CN: leaf carbon nitrogen ratio.

change the direction of general conclusions but effect sizes (except for seed mass for environmental filtering, Figure 3). Effects were more pronounced when considering abundances for competition filters of the integrated trait and seed mass and for environmental filters of the integrated trait, vegetative height. Effect sizes were larger for LDMC when the warming effect was on MDN.

DISCUSSION

Under the ongoing climate change, we require more than ever a method that can disentangle jointly acting processes in community assembly and that is sensitive to detect the onset of community re-assembly. Here, we introduced such a conceptual framework and demonstrated its efficacy. Through the unique combination of our spatially explicit trait-based framework and a reciprocal transplant experiment, we disentangled shifts in jointly acting assembly processes under climate warming. Under control conditions, signals of environmental filtering and competition were omnipresent in the functional diversity patterns of alpine and subalpine grasslands. Consistent with the stress gradient hypothesis, environmental filtering was an important structuring

force in the stressful alpine site, whereas it was much weaker in the subalpine site. Interestingly, prevailing competition processes shifted from symmetric competition at the alpine site to hierarchical competition at the subalpine site, but did not become weaker overall. Two years of experimental warming led to a shift of structuring assembly processes: Signals of environmental filtering became less important and signals of symmetric competition tended to increase, especially for drought-related traits, but no shift towards hierarchical competition was observed.

Interplay of assembly processes in mountain grasslands

The unveiled interplay of environmental and biotic signals in our study was consistent with theoretical expectations Chalmandrier et al., 2013, as well as demographic studies Hülber et al., 2011 and on-site observations in alpine ecosystems Klanderud et al., 2015. We found strong signals of environmental filtering in alpine communities which is a common observation in stressful alpine conditions where short growing seasons, low number of growing degree days and nutrient limitation impose strong restrictions to successful functional

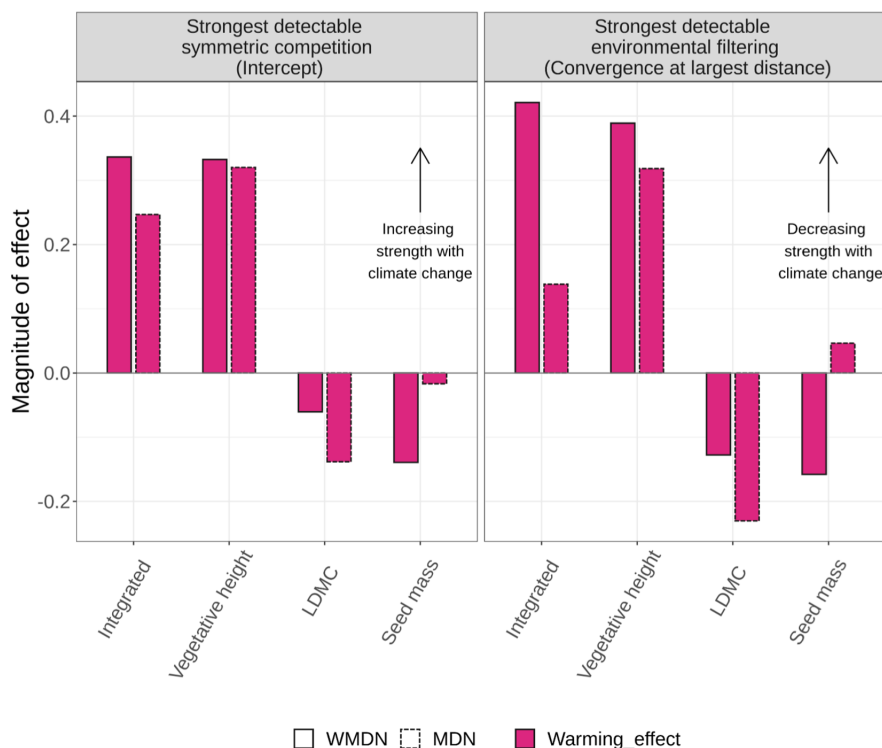


FIGURE 3 Effect of warming on the regression indices of the integrated trait, vegetative height and LDMC, i.e., when signals of filters were significant or with strong tendency in AlpineWarmed communities (Figure 2). Warming effects are estimated as the difference of regression indices between AlpineWarmed and AlpineControl communities, either under the consideration of abundances (WMDN) or only under consideration of presence/absences (MDN). (a) The intercept of the regression as the strongest detectable signal of competition and (b) the regression-based estimate of SES (see Box 1 and Figure 2) at the largest studied distance as the strongest detectable environmental filtering. Changes in indices can only be interpreted in combination with the knowledge of the original values. (a) Positive intercept differences mean that intercepts moved closer to zero which could indicate increasing symmetric competition with climate change (and vice versa). (b) Positive differences of functional distances at largest spatial distance mean that these values moved closer to zero which could indicate a decrease in environmental filtering with climate change.

strategies (Chalmandrier et al., 2017; Martinez-Almoyna et al., 2020; Midolo et al., 2021; Weil et al., 2021). In contrast, in the more benign subalpine conditions, we only detected a signal of environmental filtering for LDMC. The filtering pressure on this drought-related trait is in concordance with our observation that the subalpine soil was more water-limited than the alpine site during the growing season, potentially due to increased evapotranspiration (Figure S13).

Signals of biotic interactions also differed between the two sites. In the alpine site, we found signals of symmetric competition. While some previous work suggests that facilitation can replace competitive interactions in stressful conditions (Kikvidze et al., 2005; Michalet et al., 2006), many other studies confirm our result that competitive interactions can prevail in alpine environments (Hülber et al., 2011; Klanderud, 2010; Zhou et al., 2019). In contrast to the stress-gradient hypothesis, we did not find an overall higher competition strength in the more benign subalpine site, but rather a shift to a mixture of symmetric and hierarchical competition. The higher energy input in the still constraining subalpine site may increase niche width around the best-adapted strategy and may promote coexistence of similar species

instead of species with different niches. Due to higher energy input, subalpine communities harbour more species than alpine communities, and it has been shown that invasion of open niches is easier in species-poor communities while invasion by look-alikes is easier in species-saturated communities (Scheffer & van Nes, 2006). Moreover, Hillebrand (2019) argues that more strongly dominated communities, like subalpine communities, are also more strongly characterized by **asymmetric competition than more even communities, like alpine communities**. It would be interesting to further investigate how types of competition (and not strength of competition) change along stress gradients.

Warming induced community reassembly

We warmed perennial alpine grassland communities that are adapted to strong intra- and inter-annual climatic variability. Thus, we expected only moderate responses to warming after only 2 years of treatment. Nevertheless, with our framework, we were able to detect even these small shifts in the relative importance of community assembly processes. As expected, we found strong evidence

for a release from environmental filtering and a very weak increase of competition strength for the integrated trait and vegetative height. With warming, a new signal of symmetric competition and environmental filtering appeared for LDMC. This trait is tightly linked to drought tolerance of species and may indicate that water is becoming a limiting resource for which plants compete at the subalpine site (Figure S13). While along the stress gradient a shift from symmetric to hierarchical competition was evident, we did not observe any signals of hierarchical competition in alpine warmed plots (yet). This supports our arguments from above that such shifts in competition types may be linked to species richness and dominance which both have not yet changed much for the perennial alpine communities.

As most changes of signals of assembly processes were stronger in abundance than in presence-absence-based analyses, we conclude that changing assembly processes after warming affect first abundance structures before species' colonization and extinction. This difference between abundance and presence-absence based metrics was especially strong for environmental filtering. The reason may be that warming rarely pushes alpine species out of their abiotic niche and thus does not lead to many extinctions. Congruently, earlier observational studies found an increase in abundances of thermophilic Rumpf et al., 2018 and tall Bjorkman, Myers-Smith, Elmendorf, Normand, Rüger, et al., 2018 species under increasing annual temperatures while extinctions were rarely observed Rumpf et al., 2019. As an exception, we found that signals in LDMC were more pronounced with presence-absence than abundance-based analysis. This could be an indication that warming-associated drought drives drought intolerant functional strategies quickly to extinction. For biotic interactions and in congruence with our results, a tundra transplant experiment showed that negative effects of biotic interactions can be strong and quick and lead to colonization and extinction already during the first 4 years of warming Vandvik et al., 2020.

Towards a spatially explicit trait-based community assembly framework

Our novel spatially explicit trait-based community assembly framework allowed us to uncover small shifts in the relative importance of environmental and biotic processes in community reassembly even after short-term climate warming. Simply applying the traditional filtering approach to our climate change experiment would have revealed functional convergence for most plots at all spatial scales. This could have misled to the interpretation that environmental filtering is the only structuring force in these communities. However, explicitly including the relationships between functional and spatial distances allowed not only identifying the important role of competition but also the changing importance of

environmental filtering and the shift between different types of competition. The simplicity of our framework in quantifying the relative importance of environmental and biotic processes can be used to compare joint community assembly processes along environmental gradients and to follow transient dynamics of these processes after experimental treatments in experiments.

Here, we have paved the way for future applications of this framework but some challenges still remain. First, our framework is well adapted to disentangle environmental filtering from biotic interactions and more specifically, symmetric competition from hierarchical competition. However, it can neither disentangle facilitation from competition nor hierarchical competition from small-scale environmental heterogeneity (see introduction and Box 1). For these questions, additional analyses such as co-occurrence tests (Appendix S3) and small scale environmental measures of microhabitats are required Conti et al., 2017. Second, the choice of the regional species pool depends on the research question Münkemüller et al., 2020. Our species pool was spatially and habitat wise restricted to our experimental sites and should already account for evolutionary and large distance-related processes (e.g., dispersal). Other species pools and null models are commonly used to refine assembly tests, and these methodological advances can be easily integrated in our framework. Third, we chose a linear relationship between spatial and functional distances to be able to extract easily comparable and comprehensible parameters. In reality, this relationship may not be linear, but logistic or threshold driven (Morlon et al., 2008; Thuiller et al., 2010). Our main conclusions will not be affected by this choice as we solely compare effects between different treatments of the same type of communities. However, it could be interesting in itself to better quantify the relationship between the strength of biotic interactions and the spatial distance (e.g., at which spatial distance does the signature of biotic interactions completely fade away?). Such questions have been previously studied (Adler et al., 2010; Chu & Adler, 2015; Silander Jr & Pacala, 1985), but many questions still remain Graco-Roza et al., 2022, especially in species-rich communities where different species can maintain biotic interactions at different spatial distances. While above-ground structures that influence the distance dependency of biotic interaction may be relatively easy-to-measure traits, belowground structures are much more challenging to study not only because little is known about specific root traits but also because microbial associates will play an important role in distance-decay patterns. Fourth, how much we can learn from applying this framework depends first and foremost on the trait data used in the study. Here, we used species-specific measures of aboveground traits from trait databases. If finances and time allow, it could be highly valuable to include on-field trait measurements to be able to further investigate the role of species-specific phenotypic

plasticity to warming in reshaping the communities under climate change Henn et al., 2018). Finally, we acknowledge that a massive data collection is needed for this framework. This may not be optimal in the context of global change research, especially if we would like to (i) cover large spatial scales to provide representative results, (ii) have multiple repetitions during or between years to capture transient responses relevant for society and (iii) integrate plastic responses of species traits to account for ecosystem adaptability. However, a possible solution for this may be community composition tracking with fine resolution hyperspectral imagery (Sarić et al., 2022; Thomson et al., 2021; Wang et al., 2020). From these images, we can potentially extract the information of in situ functional information and spatial distances between individuals and can compare many communities over space and time.

CONCLUSIONS

Through our spatially explicit trait-based approach, we paved the way for a novel framework in community ecology that addresses simultaneously many major issues of the original framework of assembly processes. We used the basic but fundamental knowledge that species coexistence depends not only on their functional but also on spatial distance to unravel the processes acting together in plant communities. The unique combination of our framework with a reciprocal community transplant experiment in the French Alps allowed us to test the framework on assembly processes in mountain grasslands and detect community restructuring under climate change. Our framework has shown to be useful and sensitive enough to capture reassembly dynamics that affect primarily abundance structure, and less species turnover. We believe that the simplicity and flexibility of our framework will allow future studies to answer theoretical questions in community ecology and to better understand the effects of global change on community restructuring.

AUTHOR CONTRIBUTIONS

B.B., W.T. and T.M. conceived the ideas; B.B. and T.M. designed the methodology; B.B. analysed the data and led the writing of the manuscript together with T.M. and W.T.; B.B., M.-P.C., J.R., M.G. and I.C.S. realized the botanical surveys; J.R. managed the database. All authors gave final approval for publication.

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SUPPORTING INFORMATION

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